

Pipelines to Insights – Project Proposal

Louisville Golf Tee Time Weather Advisor

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Target Audience: Golfers in Louisville, Kentucky

Introduction

Golfers frequently struggle with deciding the best time to book tee times because weather conditions can change rapidly throughout the day. Rain delays, thunderstorms, extreme heat, and high winds can negatively impact the golfing experience and often lead to cancelled rounds or unfinished games.

Louisville, Kentucky has many popular golf courses that are heavily influenced by changing weather patterns, especially during spring and summer months (If you can't tell this has been inspired by the recent weather). A weather-focused golf application could help players make smarter scheduling decisions by analyzing forecast data and providing useful recommendations.

This project proposes the development of an interactive golf weather dashboard that uses weather forecast data from the Open-Meteo API to help golfers find the best tee times at Louisville-area golf courses. The application will provide golf specific weather information, tee time cancellation probability estimates, and intelligent recommendations such as whether golfers should play 9 holes instead of 18 due to incoming storms.

Problem Statement

Golfers often rely on basic weather apps that do not provide golf-specific recommendations or risk analysis for tee times. Players may begin a round only to encounter rain delays, lightning, or unsafe conditions midway through play. Our app hopes to help with knowing the following.

- The likelihood of a tee time being interrupted or cancelled due to weather
- Whether conditions are better suited for 9-hole or 18-hole rounds
- Which time slots throughout the day provide the safest and most enjoyable golfing experience

This project aims to solve this problem by creating a weather-driven golf recommendation system using forecast data and predictive logic.

Objectives

- Extract hourly daily weather forecast data from the Open-Meteo API using Python
- Transform weather JSON responses into structured and analyzable datasets
- Estimate the probability of tee time cancellations due to rain or storms
- Recommend ideal tee time windows for Louisville golf courses
- Suggest 9-hole versus 18-hole rounds based on forecast timing
- Store cleaned forecast data in CSV files
- Build an interactive dashboard using Dash and Plotly
- Display visual weather trends and golf recommendations through charts and KPIs

Technical Approach

Weather dashboard and Interactive UI

- I envision what looks like a weather app with real time weather displayed and a forecast for the coming days
- Ability to filter the day and the time of day, the golf course you want to play at, available tee times at these locations with relevant information and a 'rating' of how good the weather is to play golf on that day/time at the specific course
- Risk percentage of chances of cancellation of tee time. Often golf courses/booking apps will just show you the tee time anyway even if it will almost certainly be cancelled
- Suggested best available tee times based on weather factors, recommendations of where you could play and maybe sneak in a clear 9 hole window on bad weather days, advising you to not pay for 18 holes only to be rained out half way through

Primary API: Open-Meteo API

- Endpoint: <https://api.open-meteo.com/v1/forecast>
- Location: Louisville, Kentucky
- Forecast Type: Hourly and Daily Weather Forecasts

Weather Parameters

- Temperature
- Precipitation probability
- Rain amount
- Thunderstorm risk
- Wind speed

- Humidity
- Cloud cover

Weather Parameter Analysis

- A rating system that uses certain weather elements to give a score out of 100 of how good the weather is for golf
- Example 100 day could be 70 degrees, 0% chance of rain, 0mph wind, 0% thunderstorm risk all day
- Example 50/100 could be 75 degrees, 45% chance of rain, 15mph winds, 30% chance of thunderstorms

ETL Workflow Design

1. Extract

- Use Python requests library to call the Open-Meteo API
- Pull hourly forecast data for Louisville weather conditions
- Retrieve forecast updates daily or every few hours

2. Transform

- Convert JSON API responses into structured Pandas DataFrames
- Clean missing or inconsistent weather values
- Create calculated fields including:
 - Rain Cancellation Probability
 - Recommended Tee Time Window
 - 9-Hole vs 18-Hole Recommendation

Example Recommendation Logic

- Low rain probability + mild temperature = Excellent golfing conditions
- Thunderstorms expected within 3 hours = Recommend 9 holes
- Heavy rain probability >70% = High cancellation risk
- Wind speeds >20 mph = Poor golfing conditions

3. Load

- Store cleaned datasets as CSV files
- Maintain historical forecast data for visualization and analysis with option to make changes if certain conditions are deemed 'bad golf' but are okay days to play

Technology Stack

Backend

- Python
- Requests
- Pandas
- CSV storage

Frontend

- Plotly
- Dash Bootstrap Components
- Dash

Dashboard Features

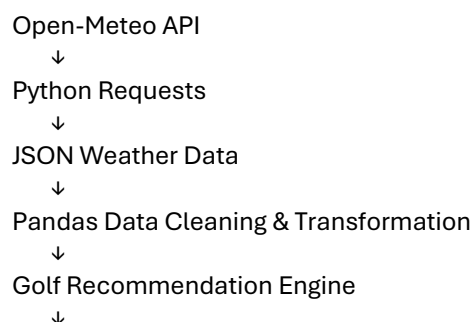
Planned Visualizations

- Hourly Weather Timeline
- Tee Time Recommendation Heatmap
- Rain Cancellation Risk Gauge
- Temperature and Wind Trend Charts
- Best Time Slots for Golfing Dashboard Cards

Planned KPIs

- Best Tee Time of the Day
- Lowest Rain Risk Time
- Golf Playability Score
- Estimated Round Completion Probability

ETL Architecture Diagram



CSV Storage



Dash + Plotly Dashboard



User Golf Tee Time Recommendations

Timeline

Week 1: Set up Dash structure, implement Open Meteo API integration, Design pipeline

Week 2: Create interactive weather visualizations, Create Database Scheme
documentation and ER diagram + SQL script

Week 3: Transform and clean data, make scoring algorithm, implement storage layer,
add any other UI features and user personality features

Week 4: Test across devices, gather and use feedback to fix issues

Week 5: Monitor API reliability and performance, prepare for ongoing maintenance

Expected Outcomes

By the end of the project, the completed system will include:

- A fully functioning ETL pipeline using Python
- Automated retrieval of weather forecast data from Open-Meteo API
- Structured and cleaned weather datasets
- A golf weather recommendation engine
- Interactive dashboard visualizations in Dash
- Tee time recommendation analytics for Louisville golfers
- Cancellation probability estimates based on weather forecasts